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Measuring Teachers' competencies for AI integration: Development and validation of the AI-TPACK in vocational education

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ABSTRACT

Integrating artificial intelligence (AI) into education necessitates teachers acquiring competencies aligned with technological advancements, especially within vocational contexts. This study aimed to adapt and validate a concise self-report instrument, the AI-integrated Technological Pedagogical Content Knowledge (AI-TPACK) scale, grounded in the TPACK framework, to measure vocational teachers' competencies in integrating AI into instructional practices. A total of 460 pre-service and in-service vocational teachers from Indonesia participated. The adapted instrument encompasses seven constructs, including AI Pedagogical Knowledge, AI Content Knowledge, AI Technological Knowledge, and their intersections, culminating in a comprehensive AI-TPACK construct. Confirmatory factor analysis confirmed strong model fit, and convergent and discriminant validity, internal consistency, and composite reliability met acceptable thresholds. Structural equation modeling revealed significant predictive relationships among constructs, while measurement invariance tests supported its suitability across pre-service and in-service teachers. These findings affirm the adapted AI-TPACK scale as a reliable and valid tool for assessing AI-integrated pedagogical competencies specifically within vocational education contexts.

1. Introduction

The increasing integration of artificial intelligence (AI) into various sectors of society has prompted growing attention to the role of AI in education. As AI continues to reshape how people learn and work, educators are expected to cultivate learners' competencies for navigating and co-existing with intelligent technologies [39]. Consequently, there is a pressing need to prepare teachers to integrate AI meaningfully into instruction. Teachers' technological and pedagogical readiness is critical, especially as AI education begins to emerge across K–12 and higher education contexts [63]. However, despite the growing interest in AI for education, a significant gap remains in supporting teachers' knowledge development for AI integration, particularly in vocational education and training (VET), where emerging technologies must align with domain-specific practices. Vocational teachers require unique technological and pedagogical competencies due to the hands-on, industry-oriented nature of their teaching. Their integration of AI and other advanced technologies often involves specialized tools and practical applications not commonly found in general education. Therefore, a dedicated instrument for vocational teachers is needed to assess competencies and specific vocational education contexts.

To address this need, the Technological Pedagogical Content Knowledge (TPACK) framework [34] offers a valuable lens for examining the integration of digital technologies in teaching. TPACK identifies the interplay among technological, pedagogical, and content knowledge as essential for effective teaching with technology. The TPACK framework offers a comprehensive model for teachers to integrate technology effectively within their teaching practices [24]. TPACK

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highlights the dynamic relationship among technological, pedagogical, and content knowledge, with pedagogy playing a central role in integrating technology and content to create compelling learning experiences [8,36]. In recent years, researchers have adapted the TPACK framework to include competencies relevant to digital, computational, and AI-based tools. Integrating AI into teaching through the TPACK framework is essential to align AI technologies with pedagogical goals, ethical considerations, and the teacher's orchestrator role [6]. This approach is particularly important in vocational education, where AI can assist in technical skill development, simulation-based learning, and precision-driven tasks [17,38]. However, most existing models are in general or STEM education, with limited application in the context of AI-integrated teaching within Vocational Education and Training (VET).

This study focuses on vocational teacher education in the field of engineering drawing, an area in which 2D and 3D visualization tools, modeling, and precision technologies are central to instructional practices [53]. Engineering drawing requires accurately representing complex mechanical and automotive objects to support engineering projects [2,3,22,50–52]. AI tools offer new possibilities to automate and enhance visualization, feedback, and assessment. Given the increasing expectations for vocational teachers to adopt AI-based technologies in industry-aligned instruction, developing a framework and instrument that addresses their unique professional needs is crucial.

Against this backdrop, the present study's primary aim is to develop and validate a concise AI-TPACK measurement instrument for vocational education, rather than to evaluate national policy. We situate Indonesia only as the implementation context, while the contribution is a psychometrically validated tool applicable to VET programs facing rapid AI adoption. We therefore streamline curricular background and foreground the measurement development and validation agenda that guides the remainder of the paper. Building on the TPACK.xs model [49], the present instrument incorporates AI-related extensions across TPACK domains, such as AI-Technological Knowledge (AI-TK), AI-Pedagogical Knowledge (AI-PK), and AI-Content Knowledge (AI-CK), and their intersections. We now also explicitly reference Ning et al. (2024), who present a detailed AI-TPACK questionnaire structure and explore the relationships among knowledge elements using rigorous factor analytic and SEM approaches. While their work offers valuable theoretical and empirical insights into the hierarchical and mediating roles of core and composite AI-TPACK components, our study distinguishes itself by developing and validating an AI-TPACK measurement tool specifically tailored for the vocational education context.

Thus, this study contributes to the growing literature on AI in education by offering a contextually grounded, psychometrically validated tool for assessing AI-related teaching knowledge among pre-service and in-service vocational teachers. The findings aim to inform teacher training, curriculum design, and future research in digitally enhanced VET instruction.

1.1. AI and tpack integration

Integrating AI into the Technological Pedagogical Content Knowledge (TPACK) framework has enhanced teachers' capacity to deliver more adaptive and intelligent forms of instruction. Recent work has adapted TPACK to AI-rich teaching. Çelik [6] proposes an Intelligent-TPACK that foregrounds ethical and pedagogical integration of AI beyond mere technical use. Hava and Babayiğit [16] examine links between AI-TPACK and teachers' digital competencies, while Karataş and Ataç [21] model TPACK and AI-TPACK components via SEM. Tram [60] and Xu et al. [62] map relationships among AI-TPACK elements and their implications for instructional practice, and Runge et al. [48] show that AI-TPACK contributes to pre-service teachers' AI acceptance within a TAM framework. Collectively, these studies indicate that AI-TPACK is not a simple extension of TK, but an integrated knowledge base that shapes how teachers design, enact, and evaluate instruction with AI.

Addressing the remaining gap, this study adapts and validates a concise AI-TPACK instrument specifically for vocational education, tested with pre-service and in-service teachers in Indonesia and evaluated for measurement invariance across groups. Unlike prior work centered mainly on construct relationships in general contexts, we position engineering drawing as a representative VET testbed, yielding a more application-oriented measure that can inform curriculum design, teacher preparation, and program evaluation in TVET settings. Despite these insights, validated measurement tools to assess vocational teachers' AI-TPACK scales remain limited, particularly within Indonesia's vocational education sector.

1.2. AI-TPACK framework

The TPACK framework is rooted in Shulman's [54] concept of Pedagogical Content Knowledge (PCK), which emphasizes the integration of pedagogy and content knowledge to enhance student learning. Mishra and Koehler [34] extended PCK by adding Technological Knowledge (TK), resulting in the TPACK model, which includes Technological Content Knowledge (TCK), Technological Pedagogical Knowledge (TPK), and the combined Technological Pedagogical Content Knowledge (TPACK). As a flexible framework [35], TPACK has been adapted for various technologies and teaching contexts, such as web technologies and twenty-first-century skills [29,61]. This study focuses specifically on AI-based tools, examining TK-related components including TK, TCK, TPK, and TPACK.

TK refers to understanding the functions and limitations of technology and the skills to use it effectively [34]. TCK involves recognizing how technology interacts with specific content areas and how each shapes the other [25]. TPK relates to understanding how technology can support or hinder teaching strategies, considering both its benefits and limitations [35]. Finally, TPACK represents the integrated knowledge of technology, pedagogy, and content, allowing teachers to select appropriate tools and methods to deliver content effectively [34].

Generative AI (GenAI) refers to AI applications that use machine learning algorithms to create new content such as text, images, videos, and music based on patterns learned from large datasets, rather than being explicitly programmed [4,36,45]. Examples include ChatGPT, Bard, DALL-E, Midjourney, and DeepMind. A specific type of GenAI, large language models (LLMs), focuses on text generation and performs tasks like summarizing, translating, and analyzing data [36]. Notable LLMs such as GPT-3 and GPT-4, developed by OpenAI, are widely recognized for generating realistic and contextually appropriate responses [47].

In educational settings, AI is commonly integrated into tools such as chatbots, intelligent tutoring systems, dashboards, and automated assessment systems, which are among the most widely used AI technologies in K-12 education [1,7]. Chatbots serve as virtual agents, assisting teachers and students through text or voice interactions, supporting motivation, and providing updates on student progress [20,32]. Intelligent tutoring systems, often linked to adaptive or personalized learning platforms, offer tailored learning experiences and help reduce teachers' workload [37,40].

Based on this foundation, the present study introduces an AI-TPACK theoretical framework (see Fig. 1), which systematically extends core TPACK constructs to encompass the unique characteristics and demands of AI-based technologies in education. The detailed descriptions of the AI-TPACK elements are provided in Table 2, positioning this framework as a robust basis for measuring and analyzing teachers' competencies in AI-integrated instructional contexts.

1.3. AI implementation in the indonesian curriculum

The integration of AI into Indonesia's national education curriculum is a strategic response to the growing demands of the digital economy and the challenges of Industry 4.0 and Society 5.0. AI education is expected to strengthen students' digital literacy, computational thinking,

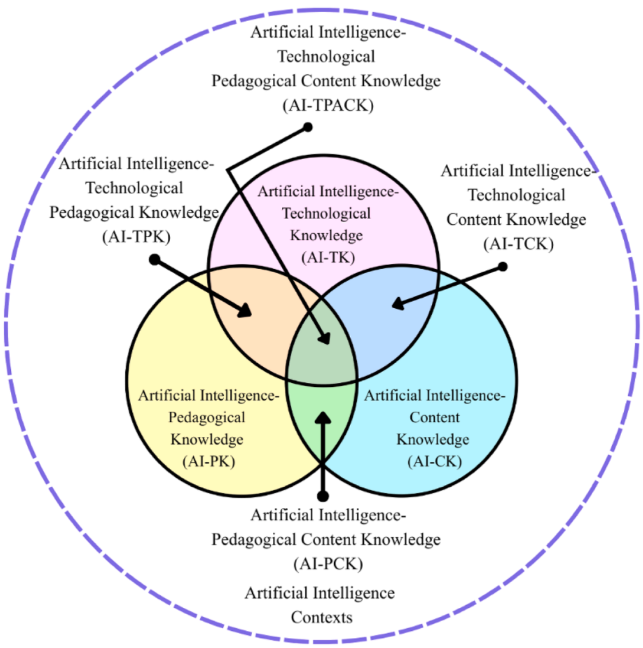

Fig. 1. AI-TPACK framework diagram.

and problem-solving abilities while fostering ethical awareness in technology use [33]. This policy aligns with international practices that emphasize AI literacy as a crucial competency for future-ready learners [17,31]. This study aligns with recent scholarship emphasizing that effective TPACK implementation must be contextually grounded [42, 43], particularly in evolving domains like AI integration in vocational education. Globally, there is increasing awareness of the importance of incorporating AI into education to drive innovation and meet the changing needs of the workforce [64]. Accordingly, Indonesia's curriculum framework explicitly identifies these competencies as essential for ensuring equitable access to quality education and preparing students to contribute to digital transformation.

Despite its progressive direction, implementing AI education in Indonesia faces practical challenges, particularly concerning disparities in infrastructure, teacher readiness, and access to learning resources. The government's strategy highlights teacher training, infrastructure provision, and multi-stakeholder collaboration as key factors for success [33]. These priorities align with existing research emphasizing the importance of adequate preparation and support for educators in adopting AI-based pedagogies [10,63]. Furthermore, scholars have argued that AI education must be contextually adapted to local educational cultures to enhance its relevance and sustainability [57]. With sustained efforts, Indonesia's policy framework aims to enhance digital competencies and position the country as an active contributor to technological innovation in education.

The successful implementation of AI education in Indonesia's curriculum requires technological infrastructure and a robust pedagogical framework, such as the integration of TPACK. Moreover, TPACK provides a comprehensive model that helps teachers effectively integrate AI technologies with pedagogical strategies and subject content to create meaningful learning experiences [23]. Integrating TPACK with AI literacy supports teachers in designing instructional practices beyond technical skills, fostering students' higher-order thinking, creativity, and ethical awareness in using AI [33]. Therefore, embedding TPACK in professional development programs for educators is critical to ensuring the sustainable and effective implementation of AI education in Indonesia's schools.

Celik [6] introduces the Intelligent-TPACK framework as a comprehensive approach to assessing and supporting teachers' readiness to integrate AI in educational settings. While the study underscores the importance of technological knowledge (TK), it also stresses that effective AI integration requires a broader range of competencies, including pedagogical and ethical understanding, to ensure responsible and meaningful implementation in the classroom. Although several recent studies—such as those by Kevser Hava and Özgür Babayiğit [16], Tram [60], Xu et al. [62], Runge et al. [48], and Karataş & Ataç [21] have explored the relationship between AI and TPACK, few have explicitly focused on a cohesive framework that integrates AI within the TPACK model.

1.4. Present study and research question

This study aims to adapt and validate a concise self-report instrument called the AI-TPACK scale to measure vocational teachers' competencies in integrating artificial intelligence within the TPACK framework. Following the principles of parsimony and practical application, the study seeks to produce a reliable and valid tool that captures all seven components of AI-TPACK while minimizing response burden. A shorter instrument improves feasibility for use in large-scale studies and helps reduce respondent fatigue, without compromising psychometric quality (Rammstedt and Beierlein, 2014; Schweizer, 2011). Accordingly, the primary research questions are outlined as follows:

1. What are the descriptive statistics of AI-TPACK among pre-service and in-service vocational teachers?
2. To what extent is the AI-TPACK instrument valid and reliable based on measurement model?
3. How is the structural assessment of the AI-TPACK construct in this study?
4. Does the AI-TPACK measurement model demonstrate measurement invariance across pre-service and in-service vocational teacher groups?

2. Research methodology

This section provides a clear and detailed explanation of the participants, data collection procedures, instrument development, and data analysis methods used in the construction of the instrument for this study.

2.1. Participants

The study involved 460 pre-service and in-service vocational education teachers in Indonesia as presented in Table 1 who participated in the Vocational Teacher Development Programme (VTDP), which focused on integrating artificial intelligence into the development of learning materials for engineering drawing courses. The VTDP invited all pre-service teachers from 13 universities offering vocational education teacher training programs. In addition, the program extended invitations to in-service vocational teachers in technology-related fields through social networks and professional communities. First, the study

Table 1
Demographics of participants.

Variables	Category	Frequency (n)	Percentage (%)
Gender	Male	365	79 %
	Female	95	21 %
Age group	<23	192	42 %
	23–33	156	34 %
	34–43	73	16 %
	44–53	31	7 %
	>53	8	2 %
Teacher status	Pre-service teacher	274	60 %
	In-service teacher	186	40 %
Living Place	Rural	269	58 %
	Urban	191	42 %

involved 460 vocational education participants, comprising both pre-service and in-service teachers in the field of vocational education. Participation was voluntary and based on informed consent. The final sample comprised 365 males (79 %) and 95 females (21 %). Participants were distributed across five age groups: under 23 years ($n = 192$; 42 %), 23–33 years ($n = 156$; 34 %), 34–43 years ($n = 73$; 16 %), 44–53 years ($n = 31$; 7 %), and over 53 years ($n = 8$; 2 %).

The teacher status distribution included 274 pre-service teachers (60 %) enrolled in teacher training programs and 186 in-service vocational teachers (40 %) actively teaching in secondary vocational schools. The sample represented a range of professional and educational experiences relevant to teaching engineering drawing, and was deemed appropriate for measuring the AI-TPACK scale in vocational education contexts.

2.2. Procedures

The Human Research Ethics Committee of Universitas Negeri Semarang, Indonesia approved protocols for this study. The study was conducted as part of a teacher development programme to integrate AI into engineering drawing instruction, specifically using 3D modeling and 3D printing technologies. The programme targeted pre-service and in-service vocational teachers and was delivered through a structured hybrid workshop combining on-site and online modalities.

2.3. Instruments

Data for this study were collected using a self-administered questionnaire distributed via Google Forms. The instrument was adapted from the TPACK.xs (Technology, Pedagogy, and Content Knowledge short version) developed by Schmid et al. [49]. The original items were modified to incorporate AI elements across all constructs to align with the context of AI integration in engineering drawing instruction. The resulting instrument consists of seven constructs representing AI domains of TPACK (see Table 2).

Subsequently, four experts specializing in educational psychology, computer science, educational technology, and English–Indonesian translation reviewed the initial version of the instrument. The educational psychologist had extensive experience in scale development and psychometric analysis, with numerous publications in reputable academic journals. The educational technologist had published multiple studies on the pedagogical applications of emerging technologies, including AI. The computer scientist brought over four years of experience in developing AI and machine learning algorithms. Additionally, the language expert had over a decade of experience in English–Indonesian translation and English language instruction in Indonesian educational institutions. Based on the experts' feedback, 42 items were revised for clarity and content alignment.

The questionnaire comprised three sections. The first collected demographic data, including gender, age, status of teacher, and living place. The second section included 42 items across the seven AI-TPACK scale constructs, with responses measured on a 7-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree). The instrument was translated into Bahasa Indonesia using a rigorous back-and-forth translation procedure to ensure linguistic accuracy. The complete questionnaire can be found in Appendix 1.

2.4. Data analysis

Data analysis was conducted using IBM SPSS Statistics version 30 and Mplus version 8.6. Descriptive statistics were used to summarize participant demographics, including age, gender, and area of engineering specialization. The normality of the dataset was assessed by examining skewness and kurtosis, with values between −3 and +3 indicating acceptable distribution for psychometric analysis [15].

Structural equation modeling (SEM) was performed in Mplus version 8.6 to evaluate both the measurement and structural models of the AI-TPACK framework, consisting of seven latent constructs: AI-PK, AI-CK, AI-TK, AI-PCK, AI-TPK, AI-TCK, and AI-TPACK. Confirmatory factor analysis (CFA) was first applied to verify item reliability and construct validity. Items with standardized factor loadings below 0.50 were removed. Model fit was evaluated using several indices with established thresholds: CFI and TLI ≥ 0.90, RMSEA ≤ 0.08, and SRMR ≤ 0.08 [19]. Construct reliability was assessed through Cronbach's alpha and composite reliability (CR), with values ≥ 0.70 considered acceptable [58]. Convergent validity was confirmed using average variance extracted (AVE ≥ 0.50), while discriminant validity was assessed using the Fornell-Larcker criterion.

Measurement invariance testing was conducted to determine whether the AI-TPACK scale measurement model was equivalent across different teacher groups (pre-service and in-service). This process followed a hierarchical approach, beginning with the configural model (baseline), which allowed all parameters to vary freely across groups to examine whether the same factor structure held. Next, metric invariance was tested by constraining factor loadings to be equal, followed by scalar invariance, where both factor loadings and item intercepts were constrained. Model fit was evaluated using chi-square difference tests and CFI and RMSEA values changes between successive models. Based on the guidelines from Parada [41], invariance is supported if the change in CFI (ΔCFI) is ≤ 0.01 and the change in RMSEA (ΔRMSEA) is ≤ 0.015. The results showed minimal changes in RMSEA and CFI between the configural, metric, and scalar models, supporting the measurement invariance of the AI-TPACK scale model across pre-service and in-service vocational teachers. This confirms that comparisons of latent means and structural relationships between these groups are statistically appropriate.

Table 2
AI-TPACK constructs with concise definitions and sample items.

Construct (Code)	Definition	Sample Item
AI-Pedagogical Knowledge (AI-PK)	Apply pedagogical strategies with AI to support diverse learners, adapt instruction, assess learning, and enhance engagement.	*"I can adapt my teaching using AI tools based on what students currently understand or do not understand."*
AI-Content Knowledge (AI-CK)	Understand how AI intersects with subject concepts, practices, and developments to explore, explain, and deepen disciplinary knowledge.	*"I have sufficient knowledge about how AI relates to my teaching subject."*
AI-Technological Knowledge (AI-TK)	Know and comfortably use AI technologies relevant to education; explore, troubleshoot, and adopt innovations.	*"I frequently explore and experiment with AI tools in educational settings."*
AI-Pedagogical Content Knowledge (AI-PCK)	Design AI-supported teaching strategies tailored to subject content; use AI for evaluation and feedback.	*"I can use AI-based tools to evaluate student performance in my subject."*
AI-Technological Pedagogical Knowledge (AI-TPK)	Select and adapt AI technologies to align with effective teaching methods, learner needs, and outcomes; use AI critically.	*"I can choose AI tools that support and enhance teaching strategies."*
AI-Technological Content Knowledge (AI-TCK)	Know how AI tools and developments influence subject content/practices; use AI to explore, analyze, and present disciplinary knowledge.	*"I know which emerging AI technologies are impacting my subject."*
AI-TPACK (Integrated Knowledge)	Integrate pedagogical, content, and technological knowledge for effective, ethical, and meaningful AI-enhanced teaching.	*"I can design lessons that meaningfully combine my subject, teaching methods, and AI tools."*

3. Results

3.1. The descriptive statistics of the ai-tpack scale among pre-service and in-service vocational teachers

The descriptive analysis revealed that in-service vocational teachers consistently reported higher mean scores across all AI-TPACK scale components than their pre-service counterparts. Specifically, in-service teachers showed the highest mean score on AI-integrated technological pedagogical knowledge (AI-TPK; $M = 5.36$, SD = 1.12), while pre-service teachers scored highest on AI-integrated technological knowledge (AI-TK; $M = 5.27$, SD = 1.00). Overall, the mean scores for both groups ranged from 5.04 to 5.36, indicating a generally positive perception of their AI-TPACK scale competencies. Standard deviations ranged between 1.00 and 1.12, suggesting moderate variability in responses. The skewness values for most subscales were negative, particularly among in-service teachers (e.g., AI-CK = –1.07), indicating that responses were skewed toward higher agreement levels. Furthermore, kurtosis values were higher in the in-service group, especially in AI-CK (2.56) and AI-TCK (2.42), suggesting a more peaked distribution and greater consensus among experienced teachers. These results imply that in-service vocational teachers tend to perceive themselves as more confident and consistent in integrating AI within their pedagogical and technological practices compared to pre-service teachers.

The data presented in Table 3 from the AI-TPACK scale mostly show a normal distribution. The skewness and kurtosis values fall within the acceptable threshold of −3 to +3, with no values exceeding these limits, indicating that the distributions are generally symmetrical or normally distribute [12]. Therefore, the dataset in this study, based on the constructs, has achieved a normal distribution.

3.2. Assessment of the measurement model

The confirmatory factor analysis (CFA) results demonstrated an acceptable model fit: $\chi^2(798) = 2420.911$, $p < .001$, RMSEA = 0.066 (90 % CI: 0.063–0.070), CFI = 0.922, TLI = 0.915, and SRMR = 0.029. These indices suggest that the measurement model reasonably represents the data. The values of RMSEA and SRMR fall within acceptable thresholds (<0.08 and <0.05, respectively), while the CFI and TLI exceed the commonly recommended cutoff of 0.90. Therefore, the model is suitable for further interpretation of the latent constructs, including AI-PK, AI-CK, AI-TK, AI-PCK, AI-TPK, AI-TCK, and AI-TPACK in the context of AI-integrated teaching competence in engineering drawing education. Fig. 2 is illustrated the CFA for measurement model, and SEM analysis for structural model is illustrated in Fig. 3.

Table 3
Descriptive statistics of the AI-TPACK scale.

Group	Variable	M	SD	Skewness	Kurtosis
PST	AI-PK	5.04	1.01	0.13	−0.36
	AI-CK	5.19	1.06	−0.43	0.58
	AI-TK	5.27	1.00	−0.33	0.16
	AI-PCK	5.13	1.08	−0.27	0.12
	AI-TPK	5.23	1.09	−0.61	1.38
	AI-TCK	5.17	1.08	−0.61	1.69
	AI-TPACK	5.13	1.07	−0.64	1.38
IST	AI-PK	5.20	1.00	−0.53	0.42
	AI-CK	5.28	1.03	−1.07	2.56
	AI-TK	5.27	1.09	−0.84	1.46
	AI-PCK	5.22	1.12	−0.80	1.30
	AI-TPK	5.36	1.12	−0.90	1.57
	AI-TCK	5.33	1.07	−1.06	2.42
	AI-TPACK	5.33	1.08	−1.04	2.16

Note: PST= Pre-Service Teachers; IST= In-Service Teachers; *M*=Mean; SD=Standar Deviation.

3.3. Convergent validity

As shown in Table 3, the standardized factor loadings for all items across the AI-TPACK scale constructs exceeded the commonly recommended threshold of 0.60 [15], with values ranging from 0.75 to 0.90, indicating strong item reliability. Convergent validity was evaluated using Average Variance Extracted (AVE), while internal consistency was assessed using Composite Reliability (CR) and Cronbach's alpha. According to Teo and Noyes [59], AVE values above 0.50 and CR and α values above 0.70 indicate acceptable construct validity and reliability. All constructs met these criteria. Specifically, AI-PK (AVE = 0.88, CR = 0.59, α = 0.93), AI-CK (AVE = 0.69, CR = 0.93, α = 0.93), AI-TK (AVE = 0.67, CR = 0.96, α = 0.94), AI-PCK (AVE = 0.72, CR = 0.94, α = 0.94), AI-TPK (AVE = 0.77, CR = 0.94, α = 0.94), AI-TCK (AVE = 0.73, CR = 0.94, α = 0.94), and AI-TPACK (AVE = 0.76, CR = 0.94, α = 0.94) demonstrated strong convergent validity and internal consistency. These findings confirm that the measurement model is psychometrically sound and supports using the AI-TPACK scale as a valid instrument in this study.

3.4. Discriminant validity

The correlations between the latent variables are presented in Table 5. The correlations ranged from 0.74 to 0.93, indicating moderate to high inter-construct associations. To assess discriminant validity, the square roots of the AVE values were calculated and placed along the diagonal of the correlation matrix. Following the Fornell-Larcker criterion [13], discriminant validity is established when the diagonal elements (i.e., $\sqrt{\text{AVE}}$ values) exceed the corresponding off-diagonal correlations in their respective rows and columns. As shown in Table 4, all diagonal values (AI-PK = 0.77, AI-CK = 0.83, AI-TK = 0.82, AI-PCK = 0.85, AI-TPK = 0.88, AI-TCK = 0.85, AI-TPACK = 0.87) exceed the inter-construct correlations, confirming adequate discriminant validity for all constructs included in the model.

3.5. Assessment of the structural model

The structural model demonstrated an acceptable overall fit with the data, indicated by fit indices within recommended thresholds: $\chi^2(807) = 2751.969$, $p < .001$; RMSEA = 0.072 (90 % CI = 0.069–0.075); CFI = 0.906; TLI = 0.900; and SRMR = 0.046. These values suggest that the hypothesized model aligns reasonably well with the observed data. All factor loadings for the latent variables (AI-PK, AI-CK, AI-TK, AIP-CK, AI-TPK, AI-TCK, and AI-TPACK) were statistically significant ($p < .001$), indicating that the observed indicators appropriately measure the constructs.

In the structural paths as shown in Table 6, all hypothesized relationships among the latent variables were significant ($p < .001$). AIPK significantly predicted AI-PCK (β = 0.305) and AI-TPK (β = 0.357), while AI-CK had strong effects on AI-PCK (β = 0.748) and AI-TCK (β = 0.682). AI-TK positively influenced AI-TPK (β = 0.690) and AI-TCK (β = 0.251). These mediating constructs (AI-PCK, AI-TPK, and AI-TCK) significantly predicted AI-TPACK (β = 0.216, 0.255, and 0.528, respectively), highlighting their crucial role in integrating AI-related knowledge into complete TPACK competence. The results confirm that the three foundational domains AI pedagogical (AI-PK), content (AI-CK), and technological knowledge (AI-TK) indirectly influence AI-TPACK through their respective combined knowledge domains, supporting a hierarchical structure within AI-integrated TPACK development.

3.6. Test of measurement invariance testing across teacher status

Following the recommendations of Brown [5], measurement invariance was evaluated across two groups: pre-service and in-service vocational teachers as presented in Table 7. The initial configural model, which tests whether the factor structure is consistent across

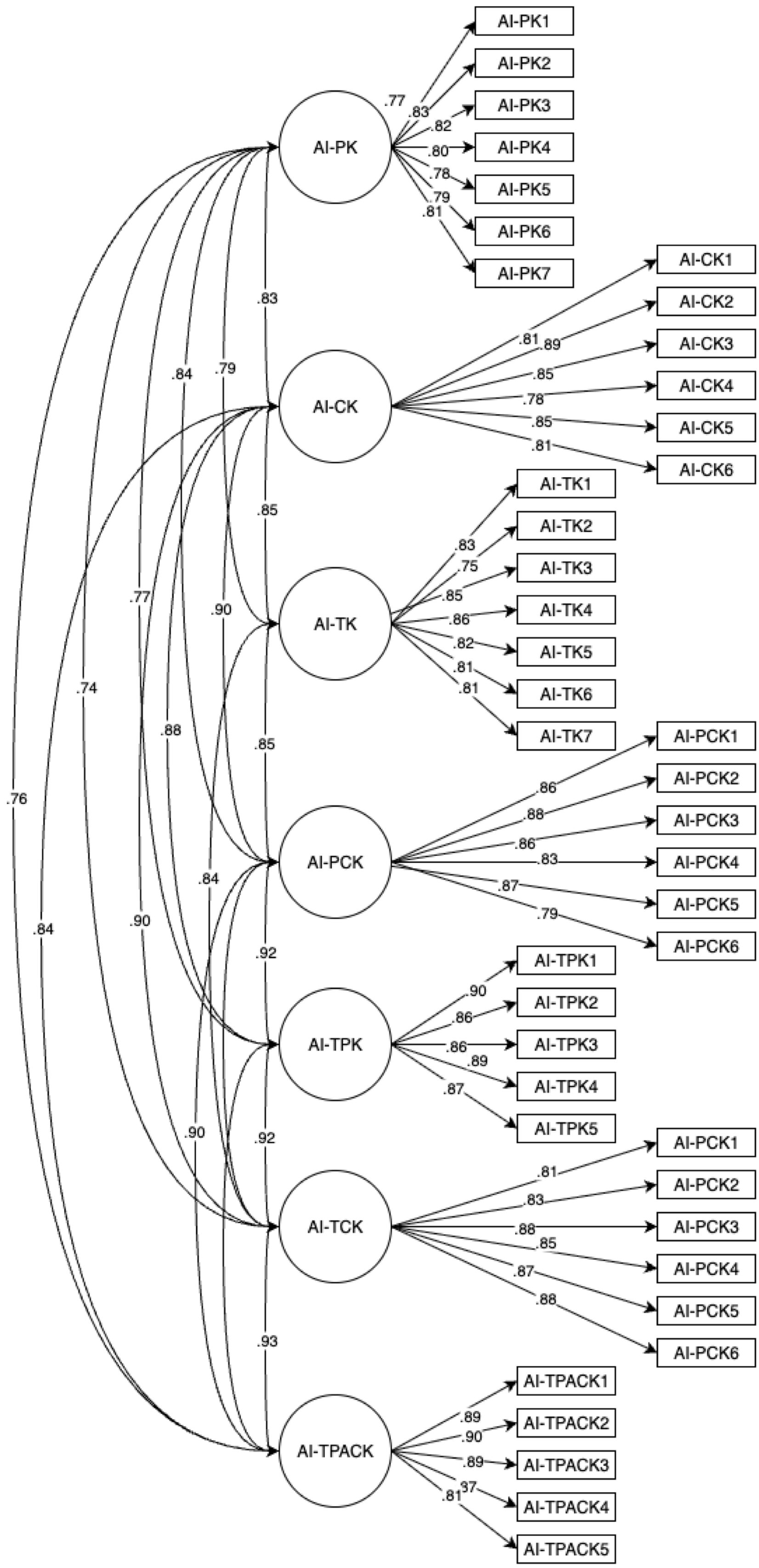

Fig. 2. The AI-TPACK scale confirmatory factor analysis reporting standardized coefficients.

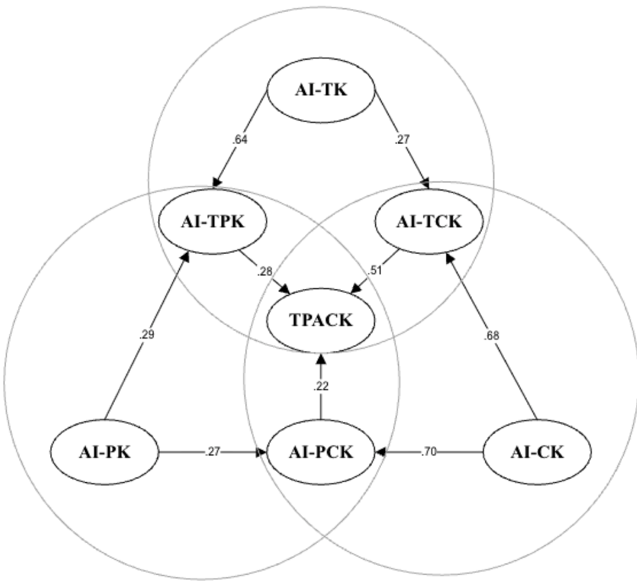

Fig. 3. Structural equation modeling AI-TPACK scale.

groups without imposing equality constraints, showed an acceptable model fit ($\chi^2 = 4015.639$, df = 1596, RMSEA = 0.081, CFI = 0.891, TLI = 0.882, SRMR = 0.037). This indicates that the AI-TPACK construct had a comparable factor structure for both groups, supporting configural invariance. The group-wise chi-square contributions were 2047.820 (pre-service) and 1967.819 (in-service), both significant at $p < .001$, confirming good model behaviour in each subgroup.

Next, metric invariance was assessed by constraining the factor loadings to be equal across the two groups. The metric model also demonstrated acceptable fit ($\chi^2 = 4065.127$, df = 1631, RMSEA = 0.081, CFI = 0.890, TLI = 0.884, SRMR = 0.045). The chi-square difference between the configural and metric models was not statistically significant ($\Delta\chi^2 = 49.489$, Δdf = 35, $p = .0531$), indicating that constraining factor loadings did not substantially worsen model fit. This supports metric invariance, suggesting that pre-service and in-service teachers attribute similar meanings to the items measuring AI-TPACK dimensions.

Lastly, scalar invariance was tested by additionally constraining item intercepts to be equal across groups. The scalar model showed acceptable fit ($\chi^2 = 4108.634$, df = 1666, RMSEA = 0.080, CFI = 0.890, TLI = 0.886, SRMR = 0.046), and the chi-square difference from the metric model was not significant ($\Delta\chi^2 = 43.506$, Δdf = 35, $p = .1532$), indicating that scalar invariance also holds. Although the comparison with the configural model ($\Delta\chi^2 = 92.995$, Δdf = 70, $p = .0345$) reached significance, the stability of CFI (no change > 0.01) and RMSEA (remained constant) suggests practical support for scalar invariance [9].

4. Discussion

This study aimed to develop and validate the AI-TPACK scale, a concise self-report instrument designed to measure vocational teachers' competencies in integrating artificial intelligence within the TPACK framework. Specifically, this study examined the descriptive competencies of the AI-TPACK scale among pre-service and in-service teachers, evaluated the validity and reliability of the AI-TPACK instrument, and assessed whether the model demonstrated measurement invariance across teacher groups. The results offer a solid foundation for interpreting these questions and demonstrate the instrument's utility in assessing AI-specific teaching competencies in vocational education.

Table 4
The standardized factor loadings, AVE, CR, and Cronbach's alpha.

Factors	Items	Standarise Loadings	Cronbach's alpha (α)
AI-PK			0.93
AVE=0.88;CR=0.59			
	AI-PK1	0.77	
	AI-PK2	0.83	
	AI-PK3	0.83	
	AI-PK4	0.80	
	AI-PK5	0.78	
	AI-PK6	0.79	
	AI-PK7	0.81	
AI-CK			0.93
AVE=0.69;CR=0.93			
	AI-CK1	0.82	
	AI-CK2	0.885	
	AI-CK3	0.85	
	AI-CK4	0.78	
	AI-CK5	0.85	
	AI-CK6	0.81	
AI-TK			0.94
AVE=0.67;CR=0.96			
	AI-TK1	0.83	
	AI-TK2	0.75	
	AI-TK3	0.85	
	AI-TK4	0.86	
	AI-TK5	0.82	
	AI-TK6	0.81	
	AI-TK7	0.81	
AI-PCK			0.94
AVE=0.72;CR=0.94			
	AI-PCK1	0.86	
	AI-PCK2	0.88	
	AI-PCK3	0.86	
	AI-PCK4	0.83	
	AI-PCK5	0.87	
	AI-PCK6	0.79	
AI-TPK			0.94
AVE=0.77;CR=0.94			
	AI-TPK1	0.90	
	AI-TPk2	0.857	
	AI-TPK3	0.864	
	AI-TPK4	0.887	
	AI-TPK5	0.874	
AI-TCK			0.94
AVE=0.73;CR=0.94			
	AI-TCK1	0.90	
	AI-TCK2	0.86	
	AI-TCK3	0.86	
	AI-TCK4	0.89	
	AI-TCK5	0.87	
	AI-TCK6		
AI-TPACK			0.94
AVE=0.76;CR=0.94		0.81	
	AI-TPACK1	0.83	
	AI-TPACK2	0.88	
	AI-TPACK3	0.85	
	AI-TPACK4	0.87	
	AI-TPACK5	0.88	

Table 5
Correlations matrix and square roots of AVEs.

	AI-PK	AI-CK	AI-TK	AI-PCK	AI-TPK	AI-TCK	AI-TPACK
AI-PK	0.77						
AI-CK	0.83	0.83					
AI-TK	0.79	0.85	0.82				
AI-PCK	0.84	0.90	0.89	0.85			
AI-TPK	0.77	0.88	0.84	0.92	0.88		
AI-TCK	0.74	0.90	0.88	0.90	0.93	0.85	
AI-TPACK	0.76	0.84	0.81	0.90	0.91	0.93	0.87

Table 6
Structural path coefficients and significance for the AI-TPACK scale.

Path	Path Coefficient	Z-Value	Result
AI-PK → AI-PCK	0.31	4.95	Support
AI-CK → AI-PCK	0.75	11.63	Support
AI-PK → AI-TPK	0.36	5.34	Support
AI-TK → AI-TPK	0.69	11.07	Support
AI-CK → AI-TCK	0.68	9.44	Support
AI-TK → AI-TCK	0.25	4.05	Support
AI-PCK → AI-TPACK	0.22	3.65	Support
AI-TPK → AI-TPACK	0.26	4.57	Support
AI-TCK → AI-TPACK	0.53	7.97	Support

Table 7
Fit statistics of models used in measurement invariance tests.

Model	Mean χ^2	df	Mean RMSEA	Mean CFI
Configural	4015.639	1596	0.081	0.891
Metric	4065.127	1631	0.081	0.89
Scalar	4108.634	1666	0.08	0.89

4.1. RQ1: descriptive insights into ai-tpack competencies among pre-service and in-service vocational teachers

The findings from the descriptive analysis demonstrate that in-service vocational teachers consistently reported higher levels of AI-TPACK competencies compared to their pre-service counterparts. This disparity suggests that practical teaching experience contributes significantly to educators' confidence and proficiency in integrating artificial intelligence within pedagogical and technological frameworks. Notably, in-service teachers rated themselves highest in AI-TPK, which aligns with previous studies indicating that teaching experience enhances the application of technology in contextually meaningful ways [11,56]. In contrast, pre-service teachers demonstrated relatively stronger perceptions in AI-TK, potentially reflecting their recent exposure to AI tools during teacher training but without the classroom experience to fully integrate them into pedagogy [18].

The skewness and kurtosis values support the interpretation that in-service teachers exhibit greater self-confidence and a more consistent internalization of AI-related teaching knowledge. The negative skewness values indicate a clustering of responses toward agreement with the scale items [14], while high kurtosis—particularly in constructs such as AI-CK and AI-TCK—suggests that experienced teachers exhibit strong consensus in their self-assessments. These statistical patterns mirror teacher technology integration literature findings, where experienced educators report more stable and self-assured technological pedagogical content knowledge. Moreover, the higher consistency among in-service teachers aligns with the notion that real-world teaching challenges foster a deeper understanding of contextualizing and applying AI tools in vocational education.

These findings underscore the importance of structured, authentic learning opportunities for pre-service teachers to bridge the theory-practice gap. Research suggests that teacher education programs must go beyond technical instruction by embedding AI integration within classroom-based practicum experiences [30]. Doing so may help pre-service teachers internalize the complex interplay of pedagogy, content, and AI technology, fostering a more balanced and confident AI-TPACK scale profile. As AI continues to shape the educational landscape, particularly in technical and vocational education and training (TVET) contexts, ensuring that pre-service teachers develop a practical, experience-based understanding of AI integration becomes essential (Kimmons et al., 2022).

Our findings highlight the distinctive nature of AI-TPACK in vocational education and training (VET). Teaching in VET is characterized by the use of industry-grade tools, strict adherence to safety and compliance standards, simulation-based practice, and the assessment of procedural and psychomotor skills. These features are less emphasized in general education settings. The higher levels of AI-TPK observed among in-service teachers reflect the context-specific integration of AI within authentic workflow tasks, such as modeling verification and providing AI-assisted feedback. This indicates that effective AI-TPACK in VET requires the alignment of technological tools with industry standards, precise tolerances, and production processes, rather than focusing solely on content delivery. As a result, our framework advances a more application-oriented perspective on AI-TPACK that sets it apart from previous general models in the literature.

4.2. RQ2: validity and reliability evidence for the ai-tpack instrument

The validation results of the AI-TPACK instrument provide strong evidence of its psychometric robustness, addressing RQ2 concerning its validity and reliability. The CFA demonstrated an acceptable model fit across all indices, with RMSEA, CFI, TLI, and SRMR values aligning with recommended thresholds [19]. These findings suggest that the AI-TPACK model is a structurally sound representation of the seven theorized domains of AI-integrated teaching competence. The CFA outcomes echo results from previous studies validating technology-enhanced pedagogical frameworks, such as the TPACK and its derivatives [26], reinforcing the model's suitability for vocational education contexts. Specifically, including AI-related constructs further expands the scope of teacher knowledge measurement in the context of emerging digital technologies.

Regarding convergent validity, the standardized factor loadings ranged from 0.75 to 0.90 across all constructs, surpassing the commonly accepted threshold of 0.60 [15]. Additionally, the AVE values for all constructs exceeded the 0.50 criterion, while both composite reliability (CR) and Cronbach's alpha values were above the 0.70 benchmark, indicating strong internal consistency. These findings affirm that the items within each AI-TPACK subscale consistently represent the underlying constructs they intend to measure. This level of convergent validity is essential for establishing construct clarity, particularly when evaluating multi-dimensional competencies such as those involving artificial intelligence in pedagogical settings [28,59]. Therefore, the AI-TPACK instrument can be considered valid and reliable for assessing teachers' knowledge integration of AI technologies in teaching and learning.

The discriminant validity of the instrument was confirmed using the Fornell-Larcker criterion, where the square roots of the AVE values were higher than the corresponding inter-construct correlations. This indicates that each construct is empirically distinct, a crucial aspect for complex theoretical frameworks such as AI-TPACK that involve overlapping domains [13]. High inter-construct correlations, ranging from 0.74 to 0.93, suggest meaningful relationships between domains while preserving construct independence. These results reinforce the argument that while AI pedagogical, content, and technological knowledge are interconnected, they contribute uniquely to the overarching AI-TPACK competence. This aligns with prior studies that advocate for nuanced, domain-specific assessments of teacher competencies in digitally enhanced environments [55].

4.3. RQ3: assessment of structural model for the ai-tpack instrument

The structural model confirmed the theoretical pathways among AI-TPACK constructs, with all hypothesized relationships showing significant path coefficients. Foundational domains AI-PK, AI-CK, and AI-TK exhibited direct and indirect effects on combined knowledge domains (AI-PCK, AI-TPK, AI-TCK), significantly predicting overall AI-TPACK competence. This hierarchical structure aligns with the original TPACK framework, validating the instrument's construct logic [25]. The model's ability to reflect a coherent and interpretable structure supports its utility in research and practical teacher development contexts, particularly as educational institutions move toward AI-integrated instruction. These findings affirm that the AI-TPACK instrument is not only statistically reliable and valid but also theoretically sound and practically relevant for vocational education.

4.4. RQ4: measurement invariance of the ai-tpack model across teacher groups

The findings from the measurement invariance analysis provide empirical support for the structural consistency of the AI-TPACK model across both pre-service and in-service vocational teacher groups. The configural invariance model yielded acceptable fit indices, which serves as a baseline by testing whether the same factor structure is valid for both groups without parameter constraints. This result indicates that participants from both groups conceptualize the AI-TPACK constructs similarly, reflecting a shared understanding of the integrated pedagogical, content, and technological knowledge required for AI-based instruction. Such configural equivalence is essential for establishing a foundation for further invariance testing [44]. These results align with previous studies that emphasized confirming consistent factor structures before interpreting between-group comparisons [27,46].

Beyond structural consistency, the analysis demonstrated metric and scalar invariance, supporting the equivalence of factor loadings and item intercepts across the two groups. The non-significant differences between the metric and configural models suggest that pre-service and in-service teachers interpret the relationships between observed items and their latent constructs similarly, enabling meaningful comparisons of their AI-TPACK levels. Scalar invariance affirms that any group differences observed in latent means are not artifacts of measurement bias but reflect genuine variations. These results support the psychometric robustness of the AI-TPACK scale and its utility in comparative studies across teacher development stages. The invariance findings reinforce earlier work suggesting that TPACK-related constructs can demonstrate broad applicability across educational levels and contexts when carefully adapted to emerging domains such as AI [26,49].

4.5. Implications of the study

4.5.1. Theoretical contributions

This study advances the theoretical landscape by extending the TPACK framework to address the integration of AI in vocational education. By developing and validating an AI-TPACK instrument specifically for vocational teachers, the research contextualizes the constructs of pedagogical, technological, and content knowledge for the distinctive, practice-based demands of VET. This work expands the applicability of TPACK to emerging digital technologies and specialized educational settings, providing new conceptual and empirical insights into the measurement tools to assess teacher competencies for AI-enhanced vocational instruction. The results offer a robust foundation of instrument validation and development for future research on VET major. Additionally, the AI-TPACK scale can be used to apply technology integration frameworks and their adaptation to rapidly evolving educational contexts, especially in vocational teacher context.

4.5.2. Practical implications

The AI-TPACK scale holds meaningful practical value for various educational stakeholders. For pre-service programs, we recommend integrating AI-TPACK through (a) design studios that pair AI tools with lesson planning, (b) micro-teaching + AI analytics for feedback cycles, and (c) school-based practica using industry-aligned tasks (CAD reviews, AI-assisted rubrics). For in-service profesional development (PD), brief micro-credentials on AI-TPK (classroom orchestration, formative assessment with AI) can consolidate existing TK. Programs should align with policy/ethics and local infrastructure to ensure feasible enactment. These steps operationalize the scale's subdomains into trainable competencies and directly target the identified gap. For policymakers, it serves as a valuable framework to identify strengths and gaps in AI competencies across vocational education, enabling thoughtful, data-informed decisions that support teacher growth and equitable access to AI-enhanced learning. For researchers, it lays the groundwork for systematically tracking professional learning outcomes and deeper exploration of AI-related teaching competencies over time. Ultimately, this scale is a step toward empowering educators and institutions to thoughtfully and effectively navigate the evolving role of AI in education.

4.6. Limitations and future directions

Despite its contributions, this study has several limitations. First, reliance on self-report data may introduce social desirability bias; future research should incorporate observational or log data to capture actual AI practice. Second, the sample focused on vocational teachers and was overrepresented in-service participants, limiting generalizability; future studies should include diverse vocational domains and include both pre-service and early-career teachers to enhance representativeness. Third, the study addressed only four common AI tools; future research should expand the instrument to include educational robots, virtual labs, and intelligent tutoring systems. Finally, the findings are culturally situated within Indonesia's VET system and teacher development pathways; generalizability may be limited where policies, infrastructure, and industry partnerships differ. Cross-cultural replications can test the instrument's transportability and inform context-sensitive norms.

5. Conclusion

This study presents the adaptation and validation of the AI-TPACK scale, aimed at assessing vocational teachers' competencies in integrating AI. Grounded in the TPACK framework, the instrument incorporates AI-specific pedagogical, content, and technological knowledge, along with their intersections, to address the increasing demand for digitally competent educators in AI-enhanced learning environments. The findings indicate that the AI-TPACK demonstrates strong psychometric properties, including satisfactory model fit, convergent and discriminant validity, internal consistency, and reliability. Structural modeling confirms significant relationships among the sub-constructs, while measurement invariance testing supports its suitability across pre-service and in-service teacher groups. These results underscore the value of the AI-TPACK framework for informing teacher professional development, curriculum design, and educational policy related to AI integration in vocational education. Future research is encouraged to examine the instrument's applicability in diverse educational contexts and its effectiveness in predicting classroom implementation outcomes.

Ethical approval

Ethical research approval was obtained from the Ethics Committee of the IRB at the Faculty of Engineering, Universitas Negeri Semarang, under Reference Number B/5181/UN37.1.5/KP.15/2025. This study is part of a broader research project in collaboration with Johannes Kepler University, the National Research and Innovation Agency (BRIN), Azerbaijan State Economic University (UNEC), Universitas Negeri Semarang, Beijing Normal University, and Comenius University Bratislava. The research was conducted in accordance with the ethical principles outlined in the Declaration of Helsinki and complies with the requirements set by the Ministry of Research and Higher Education in Indonesia. Data, including participant information, were handled with confidentiality, and the anonymity of the collected data was ensured to protect participants' privacy.

Data availability statement

Data that supports the findings of this study are available from the corresponding author, Andri Setiyawan upon reasonable request.

Disclosure statement

No potential conflict of interest was reported by the authors.

Research involving human participants - rights

The study involved human participants who voluntarily chose to participate in the research. All research participants are guaranteed privacy and anonymity.

Informed consent

For the realization of the research, permission (consent) was sought from university students and teachers at vocational schools. Participation in the study was voluntary. The data was collected, saved and analyzed anonymously.

Consent for publication

All authors have read and approved the final version of the article.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT to improve language and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

CRediT authorship contribution statement

Andri Setiyawan: Conceptualization, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Soeharto Soeharto:** Methodology, Writing – review & editing, Validation. **Tommy Tanu Wijaya:** Methodology, Validation, Writing – review & editing. **Lilla Korenova:** Methodology, Validation, Writing – review & editing. **Zsolt Lavicza:** Methodology, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Andri Setiyawan reports financial support was provided by Johannes Kepler University Linz. Andri Setiyawan reports a relationship with Austrian Agency for International Cooperation in Education and Research that includes: paid expert testimony. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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